FreePoS User Manual

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A Free and Open Source Point of Sale System for MSMEs in Fiji

Developed by

FRCS - IT Team

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1.0 Introduction

The Fiji Revenue and Customs Service (FRCS), in partnership with the FRCS IT Team developed this project to support the digital transformation of Micro, Small, and Medium Enterprises (MSMEs) across Fiji through the development of a reliable, compliant, and cost-effective Point-of-Sale (POS) system. This initiative forms part of the Industry Experience Project (IEP), aimed at equipping final-year students with real-world skills while delivering practical, high-impact solutions for national stakeholders.

In response to FRCS's requirements for greater transparency, accurate VAT reporting, and improved compliance with the VAT Monitoring System (VMS), the project team was tasked with

customizing and enhancing an open-source platform Odoo 18. The overarching goal is to develop a POS solution that can automatically apply Fiji's VAT rates, generate fiscalized receipts, synchronize sales data securely with the VMS, and streamline financial and inventory operations for MSMEs.

A comprehensive understanding of real retail operations was achieved through case-study observations at the USP Mini Supermarket. These insights informed the system requirements and guided the customization of core Odoo modules, including Point of Sale, Inventory, Invoicing, and Purchase Order. The project further involved configuring Fiji-specific tax templates, developing a Mini Chart of Accounts, integrating GTIN-based product controls, and ensuring seamless communication with the VMS through secure API interactions.

Additional enhancements were implemented to improve system usability and operational readiness, such as the integration of barcode scanning and receipt printing hardware, an essential requirement raised by the client. Furthermore, the system is being prepared for deployment using Docker to ensure consistency, portability, and ease of installation across different environments.

Overall, this project represents a significant contribution to FRCS's efforts to strengthen tax compliance and modernize business operations for MSMEs in Fiji.

2.0 Getting Started

Before using the FreePoS system, make sure the application has been fully installed and the database has been created. All steps related to installation — including setting up Docker, building the deployment environment, launching the containers, and creating the initial Odoo database — are covered in the Installation Guide.

This User Manual should only be used after installation is complete and the system is running. The sections below explain how to navigate the system, configure the POS interface, and perform daily operations such as processing sales, managing products, and handling payments.

2.1 Logging In

After the system has been installed and the initial database has been created (as described in the Installation Guide), you can access the FreePoS backend using the administrator credentials you set during the database creation step.

When you created the database in Odoo, you were asked to enter:

- Admin Email
- Admin Password
- Master Password (used only for database operations)

To log in, open your browser and go to:

```
https://pos.local:9443/
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NOTE: Please ensure you do this after completing all steps in the installation guide.

3.0 Invoicing

The invoicing module has been designed to automate all accounting processes such as taxes, journals and Charts of Account for Fiji.

3.1 Adding Taxes

1. Click on New to add new taxes.
2. Fill in the details of new taxes for it to be part of the tax list.
3. When you add a new tax, under Advanced Settings you have to select Fiji as the country in order to use the FRCS VAT group.
4. The included price should be selected as Tax Excluded.

3.2 Support for Existing and Future Tax Labels

The POS system is designed to support all current tax labels required by the local market. Additionally, the system architecture allows for the easy addition of new tax labels in the future, should the tax authority introduce new VAT categories or modify existing ones.

3.3 Tax Rounding Rules

The POS system automatically rounds the value of tax to two decimal places to ensure consistent financial accuracy.

Rounding behavior:

- If the third decimal is less than 5, the tax value is rounded down.
- If the third decimal is 5 or greater, the tax value is rounded up.

This rounding is applied to all VAT amounts printed on receipts and stored in transaction records.

3.4 Receiving Tax Rates from TaxCore (SDC / V-SDC)

The POS system receives all applicable VAT tax rates directly from the Secure Data Controller (SDC) or TaxCore as part of the configuration parameters. These tax rates are automatically synchronized and applied within the POS, ensuring that all transactions use the officially approved VAT percentages provided by the tax authority.

3.5 Exclusive Use of Tax Rates from SDC / V-SDC

The POS system does not use any tax rate other than those received from the E-SDC or V-SDC. All VAT calculations strictly follow the tax percentages provided by TaxCore, ensuring full compliance with FRCS fiscalization standards.

4.0 Using the POS System

After you have successfully carried out module installation and configuration for all modules and configured up the FRCS V-SDC settings, you may proceed to using the point of sale.

4.1 Performing a Normal Sale

5. Open the register.
6. Add products to the cart and click on Payment.
7. Choose a payment method and confirm sale.
8. TaxCore fiscalizes the transaction and a fiscal invoice is generated.

4.2 Sale Modes

The POS also supports Advance, Training and Proforma Sales. To carry out either of those sales you must first activate it.

9. Navigate to Point of Sale → open the register.
10. Click on the hamburger icon in the top right corner to activate a particular mode.

NOTE: Only one mode can be enabled at once. For example, once Advance mode has been enabled, you must disable it before you enable either Proforma or Training.

11. Once a sale mode has been activated, process a sale in the regular fashion as described in section 4.1.

4.3 Adjusting Product Quantity in the POS

Odoo POS allows the cashier to easily increase or decrease the quantity of products added to an order. The quantity can be modified directly from the numeric keypad once the quantity field is selected.

Steps to Increase or Decrease Quantity

12. Select the Product Line: tap the product in the order list, then tap on the quantity box.
13. The POS system automatically activates quantity editing mode — you do not need to press the Qty button.
14. To increase the quantity: simply press the number on the keypad. The total recalculates automatically.
15. To decrease the quantity: press the × button, then enter the number you want.

NOTE: To remove an item entirely from the cart, press the Back button. This clears the selected product from the order.

4.4 Changing the Price of a Product

The POS system allows authorized users to modify the price of a product directly from the order screen. This is useful for discounts, promotions, or manually overriding prices when required.

16. Tap on the product in the order list.
17. Press the Price button on the keypad.
18. Press the × button to clear the current price.
19. Enter the new price. The total will automatically recalculate.

4.5 Price Lookup and Rounding Rules

During price lookup, the POS system automatically applies standard rounding rules to ensure all product and service prices are displayed with two decimal places.

- If the third decimal place is less than 5, the POS rounds the price down.
- If the third decimal place is 5 or greater, the POS rounds the price up.

4.6 Selecting Products or Services

The POS system provides flexible options for selecting products or services during a sale.

Available Selection Methods

20. Search by Product Name: type the product/service name into the search bar. The POS will filter and display matching items instantly.
21. Scan a GTIN Barcode: the cashier can scan a valid GTIN (Global Trade Item Number) barcode attached to the item. The POS automatically identifies the exact product and adds it to the order.

4.7 Applying Discounts in the POS

The FreePoS system supports applying discounts directly to individual product lines during a sale.

22. Add the desired product(s) to the order.
23. Tap on the product line you want to discount.
24. Select the Discount button (percentage icon) from the action panel.
25. Enter the discount value as a percentage (e.g., enter 10 for a 10% discount).
26. The POS will automatically recalculate the line total.
27. Continue adding items or proceed to the Payment screen.

NOTE: Discounts are applied per line only; there is no global or order-level discount in this configuration.

4.8 Splitting Payments

The FreePoS allows customers to pay using multiple payment methods in a single transaction.

28. Add all desired products to the order.
29. Tap Payment to move to the payment screen.
30. Select the first payment method (e.g., Cash) and enter the amount.
31. Tap Add or press Enter to confirm the first payment line.
32. Select a second payment method (e.g., Card) and enter the remaining amount.
33. Once the total amount paid equals the order total, tap Validate to complete the transaction.

4.9 Viewing Past Orders

The Ticket Screen in the POS allows users to view all previously issued invoices and receipts. This feature works as an Electronic Journal.

Accessing the Ticket Screen:

34. Open the POS interface.
35. Click on Actions → Completed Orders.
36. The system will display a table of all past transactions.

Each order includes:

- Date & Time of the transaction
- Receipt Number and Order Number
- Cashier who processed the sale
- Total Amount
- Invoice Label (Normal Sale, Training Sale, Copy Sale, Advance Sale, Proforma Sale, Refunds)

4.10 Automatic Reference Linking for Refunds, Copy Invoices, and Linked Sales

The FreePoS system automatically manages Document Reference Numbers required for FRCS-compliant invoices. When issuing Refund, Copy, Advance, or Proforma-related sales, the system automatically links the new invoice to its corresponding original document.

4.10.1 Refund and Copy Invoices

When issuing a Refund or a Copy Invoice, the POS automatically attaches the reference of the original fiscal invoice. This ensures full traceability, compliance with FRCS refund rules, and accurate fiscal audit trails.

4.10.2 Normal Sales Linked to Advance/Proforma Sales

When finalizing an Advance Sale or converting a Proforma Invoice into a Normal Sale, the POS links the new invoice to the original document automatically. No manual reference number entry is needed.

5.0 Fiscal Invoice

A fiscal invoice is an FRCS-compliant receipt generated after a successful fiscalization request to the TaxCore V-SDC service. The POS submits the invoice payload to the V-SDC, and the fiscalized response returns mandatory data that must appear on the customer receipt.

5.2 Mandatory Fiscal Elements

Listed below are the exact elements that must be present in your fiscal receipt as per FRCS requirements.

5.2.1 Title Line

This marks the beginning of the fiscal receipt.

5.2.2 Header Data

Provided by the V-SDC as part of the fiscalized data.

5.2.3 Cashier's Identification

The name of the cashier who processed the transaction.

5.2.4 POS Number & POS Time

POS Number represents the combination of the accreditation number and product version of the accredited POS.

5.2.5 Reference Document Number

Mandatory only for Refund or Copy invoice types. The Ref No must be printed on the invoice, containing the SDC Invoice No of the original referenced invoice.

5.2.6 Invoice and Transaction Type

Identifies the sale type (e.g., NORMAL SALE, ADVANCE SALE, REFUND, etc.).

5.2.7 Line Item Details

Each product or service sold, including name, price, quantity, and total.

5.2.8 Total Purchase, Tax Items and Total Tax

Summary of total purchase amount, individual tax lines, and total tax charged.

5.2.9 Not a Fiscal Invoice

The invoice must display the message THIS IS NOT A FISCAL INVOICE on invoice types Refund, Copy and Training. This message must be written in a font size at least twice bigger than the rest of the invoice.

5.2.10 SDC Fiscal Metadata

Includes SDC Time, SDC Invoice No, and Invoice Counter.

5.2.11 Scannable QR Code

A QR code generated by the V-SDC for verification purposes.

5.2.12 Ending Title Line

Marks the conclusion of the fiscal receipt.

5.2.13 Buyer Identification

Buyer's TIN and Buyer Cost Center fields apply only to B2B (Business to Business) fiscal invoices. Since the FreePoS is intended for consumer retail transactions, these fields are not required for standard sales and have been intentionally excluded from the default POS interface and printed invoices. For B2B sales, refer to Section 9.0.

6.0 Inventory Management

6.1 Adding a New Product

37. Go to Inventory → Products → New.
38. Enter the required product details:
    - Product Name (e.g., “Bula Coffee 250 ml”)
    - Category (e.g., “Beverages”)
    - Product Type: Goods
    - GTIN: product barcode
    - Sales Price: Tax inclusive (use Fiji VAT rate 12.5% or 0%)
    - Cost Price and Vendor details
39. Select the relevant tax that will be applied to the product.
40. Click Save to register the product in your inventory.
41. Once saved, navigate to Inventory → Products to view the full product list with stock availability, sales price, and status.
42. You can also bulk-update taxes across all products by selecting Products → Tax Products from the navigation bar, choosing the products, selecting the tax rate, and clicking Confirm.

6.2 Check Product Categories

43. Go to Inventory → Configuration → Product Categories.
44. Confirm that all relevant product categories are available.
45. Assign appropriate categories when creating products. Product categories help control VAT assignment and inventory grouping.

6.3 Importing a Product/Service Catalog (CSV or Excel)

The POS catalog can be imported in bulk using a CSV or Excel file from the Inventory module.

Steps to Import Products/Services

46. Open the Inventory App from the menu.
47. Navigate to Products in the top menu.
48. Click on the gear/settings icon next to the New button and select Import records.
49. Click Upload File and select your CSV or Excel file.
50. After uploading, check that each column (e.g., Product Name, GTIN, Price, Tax, Unit of Measure) is correctly mapped to the corresponding Odoo field.
51. Click Test or Validate to check for errors. Correct any issues and re-upload if needed.
52. Once validation passes, click Import.
53. Return to Inventory → Products to confirm the newly imported products appear correctly.

7.0 Purchase Orders

The Purchase module allows users to manage supplier information, send requests for quotations (RFQs), create purchase orders, and receive goods into inventory.

7.1 How to Create Vendors

54. Navigate to Purchase → Orders → Vendors.
55. Click the New button in the top-left corner.
56. Enter the vendor's details: Vendor Name, Address, Email/Phone, and Tax ID.
57. Click Save to store the vendor profile.

NOTE: Vendor information must be created before generating purchase orders to ensure supplier names and billing details appear correctly.

7.2 How to Create a Purchase Order

58. Go to Purchase → Orders → Requests for Quotation (RFQs).
59. Click the New button.
60. Fill out the RFQ form: select the Vendor, set an Order Deadline (optional), and under the Products tab click Add a Product. Select the product, enter quantity and unit price, and confirm VAT rate.
61. Click Save to create the RFQ. The RFQ is now ready to be reviewed and sent to the vendor.

7.3 How to Confirm a Purchase Order

62. Open the RFQ you created.
63. Review product details, vendor information, and pricing.
64. Click Confirm Order at the top of the page.
65. Once confirmed, the RFQ status changes to Purchase Order, a Receipt is automatically created in the Inventory module, and the system prepares a Vendor Bill in the Invoicing module.

7.4 How to Receive Products

Step 1: Open the Purchase Order

66. Navigate to Purchase → Orders → Purchase Orders.
67. Select the confirmed Purchase Order from the list.

Step 2: Click "Receive Products"

68. Click the Receive Products button.
69. Odoo will redirect you to the Product Receipt (Picking) form, which shows all items ordered, quantities to receive, and the warehouse location.

Step 3: Validate the Receipt

70. After verifying quantities, click the Validate button.
71. Once validated, the items are added to your Inventory and the receipt status changes to Done.

8.0 Reporting

The FreePoS system provides a built-in reporting dashboard that allows supervisors and administrators to review daily, weekly, or monthly sales activity. These reports include total sales, refunds, invoice counts, payment breakdowns, and item-level details.

8.1 Creating a Report

72. Navigate to Point of Sale → Reporting → Fiscal Reports.
73. Enter the start date and end date of the report period and click Generate Report.
74. The report will display total sales, invoices, payment types breakdown, invoice counts by type, and a full list of sold items with date, time, order reference, product, quantity, unit price, tax rate, tax amount, and total.

9.0 B2B (Business-to-Business) Transactions

The FreePoS system supports Business-to-Business (B2B) fiscal invoices. When selling to a registered business, the fiscal receipt must include the buyer's FRCS TIN (Tax Identification Number) and Cost Centre, as required by FRCS fiscalization standards. This section explains how to set up a business customer and process a B2B sale correctly.

9.1 Creating a Business Customer

Before processing a B2B sale, you must create or select the business as a customer in the system.

Steps to Create a New Customer:

75. In the POS session, tap the Customer field at the bottom of the order screen, or access it via the backend by navigating to Point of Sale → Orders → Customers.
76. The customer selection screen will appear, listing all existing customers. Click the Create button (top-left) to add a new business customer.

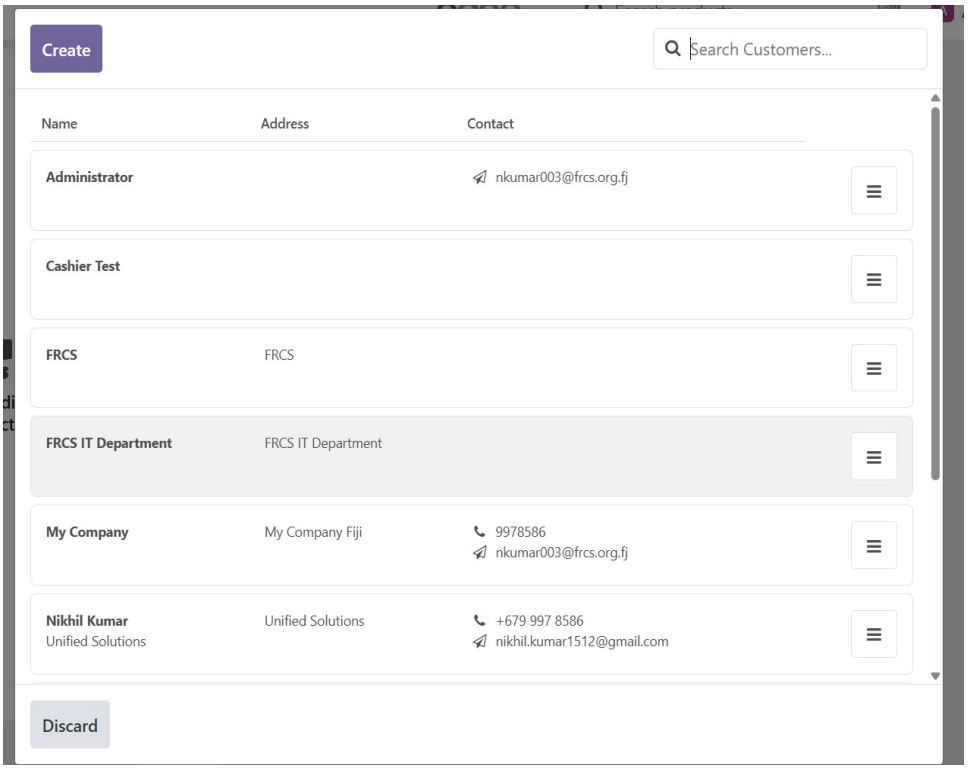

Figure 9.1 – Customer selection screen with Create button and existing customer list

77. Enter the business customer's details in the form that appears:
    - Name – the registered business name
    - Address – the business's physical or registered address
    - Contact – phone number and/or email address
78. Click Save to store the new customer record.

9.2 Entering FRCS TIN and Cost Centre

Once the customer record has been created or selected, you must enter the business's FRCS TIN and Cost Centre so that these mandatory B2B fields appear on the printed fiscal invoice.

Steps to Enter FRCS TIN and Cost Centre:

79. From the customer list, locate the business customer and click the menu icon (☰) on the right side of their row, then select Edit, or open the customer record from the backend.
80. In the Edit Partner form, scroll down to the FRCS section.

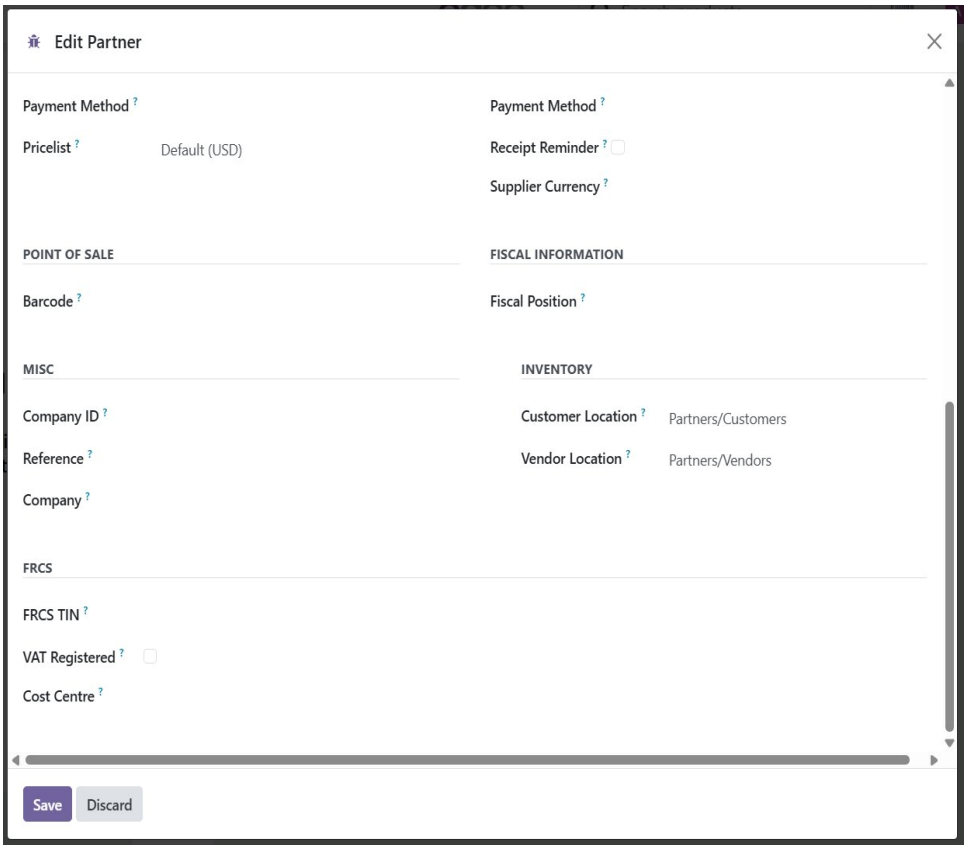

Figure 9.2 – Edit Partner form showing the FRCS section with TIN, VAT Registered, and Cost Centre fields

81. Complete the FRCS fields:
    - FRCS TIN – enter the buyer's Tax Identification Number as registered with FRCS. This field is mandatory for B2B fiscal invoices.
    - VAT Registered – check this box if the buyer is VAT-registered.
    - Cost Centre – enter the buyer's Cost Centre if applicable. This field will appear on the printed fiscal receipt alongside the TIN.
82. Click Save to apply the changes to the customer profile.

NOTE: Both FRCS TIN and Cost Centre are required for a valid B2B fiscal invoice as per FRCS requirements. If these fields are left blank, the receipt will not contain buyer identification and may be non-compliant for B2B transactions.

9.3 Processing a B2B Sale

Once the business customer has been set up with their FRCS TIN and Cost Centre, follow these steps to process a B2B sale:

83. Open the POS register (Point of Sale → Open Register).
84. Add the desired products or services to the order.
85. Tap the Customer field and search for or select the registered business customer.
86. Confirm that the correct customer is assigned to the order.

87. Tap Payment, select the payment method, and confirm the sale.
88. TaxCore fiscalizes the transaction. The generated fiscal invoice will automatically include:
    - Buyer's FRCS TIN
    - Buyer's Cost Centre
    - All standard fiscal elements (Title Line, Header Data, Line Items, Tax Summary, QR Code, etc.)

NOTE: Only one customer mode is active at a time. For B2B sales, ensure the correct business customer is selected before confirming payment. Changing the customer after payment confirmation is not possible.

9.4 B2B Fiscal Invoice Requirements

The following elements are mandatory on a B2B fiscal invoice in addition to the standard fiscal elements described in Section 5.0:

- Buyer TIN – the FRCS Tax Identification Number of the purchasing business.
- Buyer Cost Centre – the Cost Centre identifier of the purchasing business (if applicable).

These fields are automatically populated from the customer record and submitted to the V-SDC as part of the fiscalization payload. No manual input is required from the cashier at the time of sale provided the customer profile has been correctly configured.

10.0 Troubleshooting Guide

10.1 Fiji Localization Package Not Appearing in Settings

Cause:

Apps list not updated or modules not detected in custom addons.

Solution:

- Go to Apps → Update Apps List → Update.
- Ensure the l10n_fj_minicoa folder is in the custom_addons path.
- Restart the Odoo server.
- Refresh browser and check again under Invoicing → Settings → Fiscal Localization.

10.2 Fiji VAT Rates Not Showing

Cause:

Fiji package not activated.

Solution:

- Go to Invoicing → Settings.

- Under Fiscal Localization, select Fiji – Fiji – MiniCOA.
- Click Save.
- Recheck taxes under Invoicing → Configuration → Taxes.

10.3 Duplicate VAT Rates Created

Cause:

Failed installation or reinstalling without proper cleanup.

Solution:

- Uninstall the l10n module (if possible).
- Restart the server.
- Reinstall Fiji localization.
- Ensure Custom POS & POS MiniCOA are installed first (important).

10.4 Unable to Install Fiji Localization (Installation Error)

Cause:

POS MiniCOA or Custom POS not installed before l10n.

Solution:

- Install Custom POS module first.
- Then install l10n_fj_minicoa.

10.5 Journal Errors After Installation

Cause:

POS module did not create required journals.

Solution:

- Go to Invoicing → Configuration → Journals.
- Ensure Cash, Bank/Card, and Mobile Money journals exist.
- If missing: reinstall POS MiniCOA and update l10n module.